UNDERSTANDING OUR BEAMFORMING PROJECT

A simple, step-by-step guide to the Antennas and Wave Propagation report

Begin with the problem. Imagine trying to listen to one person in a crowded room. You want to hear that person clearly while reducing the voices of everyone else. A wireless receiver faces a similar problem: it can receive a wanted radio signal together with unwanted signals arriving from other directions. This project studies how an antenna array can favour the wanted direction and suppress an interfering direction.

1. The antenna idea: many elements working together

An **antenna** converts electrical signals into electromagnetic waves when transmitting, and waves into electrical signals when receiving. An **antenna array** is a group of antenna elements used together. By combining their signals carefully, the array can respond more strongly to some directions and less strongly to others.

Our report studies an **eight-element Uniform Linear Array (ULA)**. “Uniform” means the elements are equally spaced; “linear” means they lie along a straight line. The operating frequency is **2.4 GHz**, and the spacing is **62.5 mm**, half the free-space wavelength at that frequency. Angles are measured from broadside: the direction perpendicular to the line of antennas.

Essential terms

Term	Simple definition
Wavelength	Distance over which a wave completes one cycle. At 2.4 GHz it is about 125 mm in free space.
Beamforming	Combining antenna signals so the array favours a chosen direction.
Beam pattern	A graph showing how strongly the array responds at different angles.
Main lobe	The strong region pointing towards the wanted signal.
Null	A direction where the antenna response is very small.
Interference	An unwanted signal that makes reception of the wanted signal harder.

What are antenna weights? Each antenna element's signal is multiplied by a mathematical value called a **complex weight**. It controls the signal's amplitude (strength) and phase (position within its wave cycle). The eight weighted signals are combined. Correct weights can strengthen the wanted direction and reduce the interfering direction.

2. What is MVDR, and what problem are we solving?

MVDR means **Minimum Variance Distortionless Response**. It is a classical beamforming method. In simple terms, it maintains a response of one towards the wanted signal (called **unity response**) while minimizing total output power. This tends to reduce interference and receiver noise without suppressing the wanted signal.

MVDR uses a **covariance matrix**: a table of numbers describing signal powers and how signals at different antenna elements vary together. It also uses matrix calculations, including matrix inversion, to calculate the antenna weights.

The difficulty is that a very deep null depends on delicate cancellation. If the predicted weights are slightly inaccurate, the array may still point towards the wanted signal, but its interference null can become much weaker. Therefore, merely minimizing the difference between predicted and reference weights may not be enough.

Project question: Can a neural network learn to reproduce MVDR weights while preserving the deep interference null?

3. What does “physics-informed neural network” mean here?

A **neural network** is a computer model that learns a relationship from examples. It makes predictions, compares them with reference answers and adjusts its internal parameters to reduce errors. A **Physics-Informed Neural Network (PINN)** uses known physical laws or mathematical constraints to guide the model. In this project, antenna-array and MVDR mathematics are incorporated into the output construction and training objective.

The report's central idea is the **null-aligned output representation**. Instead of asking the network to predict all eight complex antenna weights independently (sixteen real values), the method first constructs an analytical vector with the required response: unity towards the desired direction and zero towards the interference direction. The network predicts a small complex coordinate called τ **(tau)**. An analytical synthesis step uses this coordinate to construct the final weights while preserving the desired-direction constraint.

Think of it this way: instead of asking a learner to guess every part of a complicated answer and hoping an important rule is satisfied, we build the answer so that the rule is automatically satisfied. The network learns the remaining coordinate that controls the interference response. This is the key contribution described in the report.

Complex number: A number with real and imaginary parts; in antenna processing it conveniently represents amplitude and phase. **Steering vector:** A mathematical list describing the phase pattern across antenna elements for a wave arriving from a particular direction. **Desired angle:** Direction of the wanted signal. **Interference angle:** Direction of the unwanted signal.

4. Dataset and training

The team generated examples using **Python** and **NumPy**. Desired and interference angles were selected on a one-degree grid from −60° to +60°. This creates 14,641 candidate pairs. The 599 pairs with angular separation below 3° were excluded, leaving **14,042 examples**. For each pair, an analytical MVDR solver calculated reference weights. These reference answers are called **ground truth**.

Data group	Number	Purpose
Training	11,233	Used to teach the network.
Validation	1,404	Used to monitor training and select a model.
Testing	1,405	Held back to evaluate performance on unseen examples.

The network is a **multilayer perceptron (MLP)**, a neural network made of connected layers. Its inputs are the two angles, scaled by dividing by 60. Hidden layers use **ReLU**, an activation function that helps learn non-linear relationships. The team used **PyTorch**, the **Adam** optimization method, learning rate 0.001, batch size 256 and up to 200 epochs. An **epoch** is one complete pass through the training data; a **batch** is a smaller group processed together.

The team compared six model variants. They differed in their output representations and **loss functions**. A loss function is a score measuring how far a prediction is from the desired result. The full null-aligned model combined an MVDR-related cost, a weight-error term and a penalty for a null that was not deep enough.

5. Understanding the results

Two important measures are used. **Null depth** measures how much the response in the interference direction is reduced relative to the desired direction. It is expressed in decibels (dB); a more negative value means stronger suppression. **SINR** means Signal-to-Interference-plus-Noise Ratio. It compares wanted-signal power with combined interference and noise power at the output. A smaller SINR gap relative to MVDR means the model performs more like the analytical reference.

Method	Median null	Meaning
Analytical MVDR	–96.17 dB	Reference result calculated using the mathematical MVDR method.
Direct neural baseline	–26.81 dB	Directly predicting all weight components did not preserve a deep null well.
Null-aligned MSE	–92.95 dB	The null-aligned representation greatly improved null preservation.
Full null-aligned model	–96.41 dB	Very close to analytical MVDR on the ideal-array test set.

The full model's median output SINR gap was approximately **0.00000543 dB** compared with MVDR. This means its output SINR was almost identical to the reference for the tested ideal-array examples. It does **not** mean a real antenna has zero interference, nor that the network is universally better than MVDR.

How to understand dB: A power ratio in dB is 10 log10(P1/P2). A very negative relative response means the interference-direction response is much smaller than the desired-direction response. Do not interpret –96 dB as “96 percent cancellation”.

6. Why did the team use HFSS?

Ansys HFSS means High-Frequency Structure Simulator. It is software used to simulate electromagnetic behaviour. The team modelled eight centre-fed dipole antennas, arranged along the x-axis, with 62.5 mm spacing. A **dipole** is a basic antenna made of two conductive arms. A **full-wave simulation** models electromagnetic field behaviour rather than relying only on a simplified ideal-array formula.

The calculated weights were applied to the simulated antenna ports. The resulting far-field patterns were compared with the ideal analytical and neural-network patterns. A **far-field pattern** shows the antenna response at observation directions sufficiently far from the antenna.

In HFSS cases 1–4, the simulated physical null depths were approximately –27.83 dB to –33.72 dB. In case 5, the desired and interfering directions were only 3° apart near endfire, and the null was –5.50 dB. These results are much shallower than the approximately –96 dB ideal-array result.

Mutual coupling is electromagnetic interaction between nearby antenna elements: one element affects the electrical behaviour of another. Port loading and other physical effects can also make a real antenna array behave differently from an ideal mathematical model. The report identifies these as plausible contributors; its five simulations do not isolate the contribution of each one.

Important distinction: –96.41 dB is the neural network's result on the ideal-array test set. HFSS gives electromagnetic simulation results for a dipole array. The report does not describe a fabricated antenna or laboratory measurements. Never claim that a physical antenna achieved a –96 dB null.

7. What the project demonstrates—and what remains

The project demonstrates that the way a neural network represents its output matters. Directly predicting all antenna weights gave a much shallower median null. The null-aligned representation builds the desired-direction constraint into the output construction and reproduces the ideal-array MVDR result much more closely. The HFSS comparison also shows that success in an ideal mathematical model does not automatically guarantee the same performance in a more realistic electromagnetic model.

The report proposes future work on multiple interferers, broadband operation, off-grid angles, different array geometries, coupling-aware training, measured calibration data and real-hardware testing. It also recommends comparing neural-network inference time with an optimized analytical solver before claiming a speed advantage.

8. Technical glossary for revision

Term	Definition	Term	Definition
Steering vector	Array phase pattern for an arrival angle.	**Complex weight**	A weight with real and imaginary parts; represents amplitude and phase.
Covariance matrix	Describes signal powers and relationships across elements.	**Ground truth**	Reference answer used to train or assess the model.
Ablation study	Comparison of model variants to test design choices.	**RMSE**	Root Mean Square Error; summarizes differences between values.
Generalization	Performance on examples not used for training.	**Mutual coupling**	Electromagnetic interaction between antenna elements.

One-minute project summary: Our project uses an eight-element antenna array to receive a wanted signal while suppressing interference. MVDR calculates reference beamforming weights. A neural network learns to predict them. The null-aligned representation builds the desired-signal constraint into the output construction and controls the interference response through a learned complex coordinate. On the ideal-array test set, the full model achieved a −96.41 dB median null, close to the −96.17 dB MVDR reference. HFSS simulations produced shallower nulls, showing why electromagnetic effects and calibration matter.

Prepared from the supplied 15-page Antennas and Wave Propagation report. The numerical claims and project details are based on that report; this guide explains them in simpler language and does not independently reproduce the simulations.

Continued on the following pages: worked example, presentation plan and viva questions.

9. What exactly did we do? A step-by-step story

Think of the project as three connected parts: an **ideal mathematical teacher**, a **neural-network learner**, and a **more realistic electromagnetic comparison**. Each part answers a different question.

Step 1 — Choose the scenario. Use an 8-element array at 2.4 GHz. Pick the desired angle and the interference angle. The study fixes desired-signal power at 100, interference power at 1000 and noise power at 1 in its normalized model.

Step 2 — Calculate the reference answer. The steering vectors describe how a wave from each angle appears across the elements. An analytical MVDR solver uses the assumed covariance to calculate eight complex reference weights.

Step 3 — Make examples. Repeat for angle pairs on the 1° grid. After excluding pairs less than 3° apart, split the 14,042 examples into training, validation and test sets.

Step 4 — Train six variants. Each network receives two normalized angles. The variants differ in how they represent the output and how their errors are penalized. Validation is used to select the saved checkpoint.

Step 5 — Test on held-out pairs. Compare predicted and analytical weights using null depth, SINR gap, pointing error and relative weight error. The test set was not used to fit the networks.

Step 6 — Compare with HFSS. Apply ideal-array weights to a simulated dipole array and compare the exported pattern with ideal MVDR and neural predictions. This is an electromagnetic simulation comparison, not a hardware experiment.

The central result: the way we construct the network output matters more than simply asking it to predict every weight independently.

What does the network actually learn?

For the null-aligned variants, the network predicts one small complex number, tau (τ): two real outputs. The final eight complex weights are then assembled using the two angles and known array equations. The analytical construction keeps the desired response at one; τ controls the interference response. The network does not independently output all 16 real weight components.

10. Understanding the design choices and results

Why six models? The ablation in plain language

Variant	What changes	Median null
Baseline	Predicts all weights directly; weight error loss.	−26.81 dB
raw16_norm	Direct weights with an added unity-response projection.	−21.45 dB
subspace_mse	Predicts a complex coordinate in a signal/interference basis; weight loss.	−26.63 dB
subspace_cost	Same subspace idea, with MVDR physics cost.	−44.72 dB
nullaligned_mse	Predicts the null-aligned coordinate; weight loss.	−92.95 dB
nullaligned_full	Null-aligned coordinate plus physics cost, weight anchor and null penalty.	−96.41 dB

These values are medians over the **1,405 held-out ideal-array test pairs**. Analytical MVDR has a median null of **−96.17 dB**. A more negative value is a deeper null. Compare subspace_mse and nullaligned_mse to see the effect of the output representation while both use weight error loss.

A small amount of mathematics, translated

Desired response = 1: the construction preserves the wanted direction. **Interference response = 8 × |τ|:** in this ideal 8-element model, its response amplitude is controlled by the magnitude of τ. Making τ small makes the null deep. An exact null (τ = 0) is not automatically the MVDR optimum, because MVDR also accounts for noise.

How to read the main numbers

Null depth: relative power in the interference direction compared with the desired direction. For power ratios, dB = 10 × log10(interference-direction power / desired-direction power). A value of −30 dB means a ratio of 1/1,000, not 30% or 30 dB of gain.

SINR gap: analytical MVDR output SINR minus model output SINR, in dB. The full model’s median gap is approximately 0.00000543 dB: very close to MVDR on these test examples. This does not prove that the network beats MVDR in general.

90th percentile: the value at or below which 90% of cases fall. For null depth, this describes the shallower tail; it is not the single worst case. Good median pointing does not guarantee a deep null.

11. A concrete HFSS example and what to show

Example: case 02. Set the desired direction to +30° and interference to −40°. The ideal MVDR calculation gives a −95.99 dB relative null at the interference angle; the network's ideal-array curve gives −99.07 dB for this case. The saved HFSS far-field export gives −33.72 dB at −40°. These are three comparisons for one geometry, not the test-set medians.

The HFSS peak is 11.27 dBi at +30°. The local HFSS notch is about −36.80 dB at −38°, so the null has shifted slightly from the requested interference angle. Coupling and port effects are plausible explanations, but the available comparisons do not separately measure the contribution of each cause.

All five simulated cases at a glance

Case	Desired / interference	HFSS null	Ideal MVDR null	What to notice
01	0° / 30°	−28.37 dB	−120.00 dB	Ideal result hits plotting floor.
02	30° / −40°	−33.72 dB	−95.99 dB	Useful worked example.
03	−25° / 45°	−32.79 dB	−95.82 dB	Shallower simulated null.
04	5° / −55°	−27.83 dB	−96.60 dB	Shallower simulated null.
05	58° / 55°	−5.50 dB	−50.85 dB	Only 3° apart, near endfire.

All nulls in the table compare the response at the interference angle with the response at the desired angle. Case 05's HFSS peak is at 82°, not 58°: nearby directions near endfire are challenging. The 181-point HFSS cut is sampled at 1° intervals.

How to read an overlay plot

The horizontal axis is angle and the vertical axis is peak-normalized response in dB. The curve's high region is the main lobe; the dip near the interference angle is the null. A curve can match the main lobe yet differ greatly at a very deep null. Whole-curve RMSE averages differences across angles; a few clipped mathematical zeros can make it large, so read the pattern as well as the number.

Do not say "our antenna achieved a −96 dB null." The approximately −96 dB result is an ideal-array test-set median. HFSS results are simulations; no fabricated antenna or lab measurement is reported.

12. A simple final-presentation plan

Use the existing report figures as visual evidence. The exact timing can be adjusted to your allotted presentation slot.

1. Motivation: Explain the crowded-room analogy: one desired signal, one interferer. Introduce beamforming, a main lobe and a null.

2. Array and reference: Show the eight-element, 2.4 GHz array and explain that MVDR calculates a reference set of complex weights.

3. Problem: Explain that deep cancellation is sensitive to tiny weight errors. A network can point accurately while failing to preserve the null.

4. Proposed idea: Show the workflow diagram. Explain the exact-null starting vector, the learned τ coordinate and the guaranteed desired-direction response.

5. Dataset and ablation: Give the total, split and six variants. Show the comparison bar chart and contrast -26.81 dB baseline with -96.41 dB null-aligned full model and -96.17 dB MVDR.

6. HFSS: Show case 02's overlay and case 05's closely spaced near-endfire geometry. Clearly distinguish ideal predictions from the full-wave simulated pattern.

7. Conclusion and future work: State what worked in the ideal model, what did not transfer directly to HFSS and why calibration and measurements are future work.

Where to find the evidence

Workflow: Figure 1 in the project report. **Model comparison:** report Table 4 and Figure 3; source: data/ablation_results.json and plots/fig2_ablation_bars.png. **HFSS cases:** report Table 5 and Figures 8–12; sources: hfss_exports/ and plots/hfss/metrics.csv. **Numerical checks:** verify_physics.py.

Important boundaries of this study

The test split is from the same integer-angle grid and fixed power assumptions as training. It does not establish performance for off-grid angles, different powers or multiple interferers. The stored results represent a recorded run, not a study over many training seeds. The saved HFSS files alone do not establish mesh convergence or isolate every physical effect. No speedup over an optimized analytical solver has been benchmarked.

Safe demonstration: Inspect the saved plots and JSON or run "python verify_physics.py" after installing requirements. Retraining or regenerating the dataset and plots can overwrite saved outputs; make a backup first. A new HFSS solve requires the Ansys software and license.

13. Viva questions: short answers you can explain

What is the project's main objective? To learn MVDR beamforming weights while retaining a deep interference null in the ideal-array model.

Why do we use multiple antennas? Their signals have different phases depending on direction. Combining them with suitable weights produces directional gain and suppression.

What is a complex weight? A value with real and imaginary components that represents both amplitude and phase; one weight is used for each element.

What is MVDR? Minimum Variance Distortionless Response: minimize output power while maintaining unity response toward the desired direction.

What does physics-informed mean here? Known array equations construct the output to satisfy the desired-response constraint, and the loss includes an MVDR cost and null penalty. Reference weights are also used for supervised learning.

Why is the null-aligned output useful? The network learns one complex coordinate controlling the ideal null, instead of relying on delicate cancellation among 16 independently predicted real weight components.

What is the difference between training and testing? Training adjusts network parameters. Validation helps select the checkpoint. The 1,405 test examples are held back for evaluation, but come from the same type of simulated angle grid.

Why compare with HFSS? The ideal model omits effects present in a simulated coupled dipole array. Comparing the patterns highlights the gap between mathematical and electromagnetic models.

Why is the HFSS null shallower? Mutual coupling, port loading and finite-array effects are plausible contributors. The stored results do not isolate or prove each individual contribution.

Why is case 05 difficult? The desired and interference angles are only 3° apart near endfire, where the array has reduced angular resolving ability and the main pattern can shift.

Is the neural network better or faster than MVDR? It closely approximates MVDR on the stated ideal test set. The project does not demonstrate a general improvement over MVDR or benchmark a speed advantage.

Was this tested on physical hardware? No. The presented electromagnetic results are HFSS simulations; a fabricated antenna and laboratory measurements are future work.

Remember the final message: strong ideal-model results are encouraging, but practical beamforming also needs calibration and physical validation.